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Research and application of deep learning-based sleep staging: Data, modeling, validation, and clinical practice

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ABSTRACT

Over the past few decades, researchers have attempted to simplify and accelerate the process of sleep stage classification through various approaches; however, only a few such approaches have gained widespread acceptance. Artificial intelligence technology, particularly deep learning, is promising for earning the trust of the sleep medicine community in automated sleep-staging systems, thus facilitating its application in clinical practice and integration into daily life. We aimed to comprehensively review the latest methods that are applying deep learning for enhancing sleep staging efficiency and accuracy. Starting from the requisite "data" for constructing deep learning algorithms, we elucidated the current landscape of this domain and summarized the fundamental modeling process, encompassing signal selection, data pre-processing, model architecture, classification tasks, and performance metrics. Furthermore, we reviewed the applications of automated sleep staging in scenarios such as sleep-disorder screening, diagnostic procedures, and health monitoring and management. Finally, we conducted an in-depth analysis and discussion of the challenges and future in intelligent sleep staging, particularly focusing on large-scale sleep datasets, interdisciplinary collaborations, and human-computer interactions.

1. Introduction

Sleep staging is the basis of sleep research and is a key step in diagnosing sleep-related disorders. However, analyzing sleep architecture is a laborious and time-consuming task requiring sleep experts to visually inspect physiological signals recorded throughout the night. Not only does this tedious and lengthy task require a high level of expertise owing to the difficulty in interpretation, but it is also susceptible to signal quality and subjective factors. Consequently, over the past few decades, there have been continuous efforts to explore methods that simplify and expedite sleep staging.

Several commercial software solutions have introduced automatic scoring services. However, owing to their considerable inaccuracies [1], the average manual correction rate is 22.2 %, which can increase to as high as 53.3 % in cases with significant signal artifacts [2]. Therefore, sleep experts consider automatic staging results inconclusive and rely on visual scoring procedures. Paradoxically, owing to the shortage of sleep physicians, sleep staging is highly reliant on automatic scoring systems in underdeveloped areas, resulting in long-standing deficiencies in diagnosing and treating sleep disorders within these areas.

With the development of the Internet, expansion of digital information, advancements in theoretical algorithms, and improvements in hardware platforms, artificial intelligence (AI) has made significant progress in the field of healthcare [3], which has a positive impact on sleep medicine. Deep learning (DL) is a representative AI technique that excels in uncovering intricate features concealed within raw physiological signals [4]. DL enables rapid automated analysis of sleep architecture, leading to significant time and economic savings, and is satisfactorily accurate. Although most studies still focus on the

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Abbreviations

AASM	American Academy of Sleep Medicine
Acc	accuracy
AHI	apnea–hypopnea index
AI	artificial intelligence
CNN	convolutional neural network
CRNN	convolutional recurrent neural network
CSTs	consumer sleep technologies
DL	deep learning
ECG	electrocardiogram
EEG	electroencephalogram
EMG	electromyogram
EOG	electooculogram
FL	federated learning
FN	false negative
FP	false positive
GRU	gated recurrent unit
HRV	heart rate variability
HSAT	home sleep apnea test
ICC	intraclass correlation coefficient
LSTM	long short-term memory
ML	machine learning
NREM	non-rapid eye movement
NT1	narcolepsy type 1
OSA	obstructive sleep apnea
PCA	principal component analysis
PD	Parkinson's disease
PPV	positive predictive value
PSG	polysomnography
R&K	Rechtschaffen and Kales
RBD	REM sleep behavior disorder
REM	rapid eye movement
RNN	recurrent neural network
Spc	specificity
STFT	short-time Fourier transform
TL	transfer learning
TN	true negative
TP	true positive
TST	total sleep time
W	wakefulness
WT	wavelet transform
κ	Cohen's kappa coefficient

methodology, with the feasibility, effectiveness, and safety yet to be assessed in real-world settings, the potential of constructing DL models to develop precise, efficient, and stable sleep-staging systems is foreseeable.

Hence, this review presents an overview of the data preparation and fundamental processes for constructing DL algorithms, along with potential application scenarios, and discusses the current limitations and future research trends in this field. The search strategy is also present here. We performed searches in PubMed to identify the relevant studies published in recent years, especially in the latest three years. The following strategy was used for searching: *(" sleep stag*" [All fields] OR "automated sleep staging" [All fields] OR "sleep scoring" [All fields]) AND ("Deep Learning" [Mesh] OR "Neural Networks, Computer" [Mesh] OR "deep neural network*" [All fields] OR "convolutional neural network*" [All fields] OR "recurrent neural network*" [All fields])*. Additionally, we consulted Google Scholar to supplement the literature, ensuring a more comprehensive list of our references.

2. Technology overview

AI is a collective term for theories, methods, technologies, and applications capable of simulating, extending, and expanding human intelligence, which is an interdisciplinary scientific field based on computer science. Machine learning (ML) is a subfield of AI as well as currently the most used method for achieving AI. ML investigates how computers simulate or emulate human learning behaviors, acquiring new features and patterns to reorganize existing knowledge structures and thereby improving itself continually. ML is mainly divided into two major categories: one is traditional algorithms based on prior knowledge, and the other is DL, which has risen rapidly in recent years [5]. Others, such as reinforcement learning, transfer learning (TL), and federated learning (FL), belong to different ML methods developed for various goals. For example, TL is proposed to use knowledge learned from "original domain labeled data" to guide the classification of "target domain unlabeled data," thereby saving time on manual labeling of samples [6]. FL, on the other hand, is essentially a secure distributed ML technology designed to conduct efficient ML among multiple participants or computing nodes while ensuring information security and protecting data privacy [7].

3. Data source

In this review, "data" refers to physiological signals obtained from sleep monitoring. These data can be collected by researchers and their collaborators, or retrieved from publicly available databases. Subsequently, these data are divided into training and validation sets for DL model development and internal and external test sets for performance assessment, as required. Table 1 comprehensively summarizes 10 frequently used databases in automatic sleep scoring research [8–17]. For each database, we present specific details, including the participants' demographics, signal modalities and channels, and relevant website links.

Two rules for visual sleep scoring directly determine the content of annotations provided by the database. According to the Rechtschaffen and Kales (R&K) rule [18], proposed in 1968, sleep is divided into wakefulness (W), non-rapid eye movement (NREM) sleep, rapid-eye movement (REM) sleep, and movement time. NREM is divided into stages 1 (S1), 2 (S2), 3 (S3), and 4 (S4), with only a central lead electroencephalogram (EEG) used for interpretation. In 2007, the American Academy of Sleep Medicine (AASM) set forth *The AASM Manual for the Scoring of Sleep and Associated Events*, which divided sleep into five stages, including W, NREM 1 (N1), NREM 2 (N2), NREM 3 (N3), and REM [19]. According to the AASM rules, EEG should be named and positioned following the international 10–20 system, with each epoch comprising 30 s for sleep-staging interpretation. In addition to the central lead EEG, frontal and occipital leads are also employed in sleep scoring to enhance the accuracy of sleep-stage classification. It is worth mentioning that the AASM manual has updated continuously to incorporate the latest and most relevant evidence in sleep medicine. The most recent update, Version 3, was released in February 2023 [20]. As new technology, automatic sleep staging had not been included in the new AASM recommendations. Therefore, we hope that our review can contribute new literature suggests.

4. Modeling approach

The construction of a DL model typically involves five steps: signal selection, data pre-processing, model architecture construction, classification task determination, and performance evaluation (Fig. 1). In different studies, there were substantial variations in these steps.

Table 1
Summary of sleep datasets from which the signals were extracted for deep learning.

Datasets	Sub-dataset	Rules	Participants & Age	Records	Signals (channels)	Website
CAP [8]	–	R&K	Mixed; 14–82, 45 ± 19	108	EEG (Fp1-F3, F3–C3, C3–P3, P3–O1 and/or Fp2-F4, F4–C4, C4–P4, P4–O2), EOG (2 channels), EMG (submentalis, bilateral anterior tibialis), ECG (1 channel)	https://physionet.org/content/capslpdb/1.0.0/
CinC2018 [9]	–	AASM	Patients; 18–93, 55 ± 14	1983	EEG (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1), EOG (left), EMG (Chin), ECG (1 channel)	https://physionet.org/content/challenge-2018/1.0.0/
ISRUC [10]	S1	AASM	Patients; 20–85, 51 ± 16	100	EEG (C3-A2, C4-A1, F3-A2, F4-A1, O1-A2, O2-A1), EOG (LOC-A2, ROC-A1), EMG (Chin, leg-1, leg-2), ECG (1 channel)	https://sleeptight.isr.uc.pt/?page_id=48
	S2		Patients; 26–79, 4.87 ± 18.7	16		
	S3		Healthy; 30–58, 40 ± 10	10		
MASS [11]	SS1	AASM	Patients; 55–76, 63 ± 5.3	53	EEG (C3, C4, Cz, F3, F4, F7, F8, O1, O2, P3, P4, Pz, T3, T4, T5, T6, Fz, and in some cases Fp1, Fp2), EOG (left, right), EMG (anterior tibialis right, anterior tibialis left, Chin1-3), ECG (1 channel)	http://www.ceams-carsm.ca/en/MASS
	SS2	R&K	Healthy; 18–33, 23.6 ± 3.7	19	EEG (C3, C4, Cz, F3, F4, F7, F8, O1, O2, P3, P4, Pz, T3, T4, T5, T6, Fpz, Fp1, Fp2), EOG (left, right, up, down), EMG (Chin), ECG (1 channel)	
	SS3	AASM	Healthy; 20–69, 42.5 ± 18.9	62	EEG (C3, C4, Cz, F3, F4, F7, F8, O1, O2, P3, P4, Pz, T3, T4, T5, T6, Fpz, Fp1, Fp2, Oz), EOG (left, right), EMG (Chin1-3), ECG (1 channel)	
	SS4	R&K	Healthy; 18–35, 25.3 ± 4.3	40	EEG (C3, C4, O1, O2), EOG (left, right, up, down), EMG (Chin), ECG (3 channels)	
	SS5		Healthy; 20–59, 25 ± 7.4	26	EEG (C3, C4, Cz, F3, F4, F7, F8, O1, O2, P3, P4, Pz, T3, T4, T5, T6, Fpz, Fp1, Fp2, Oz), EOG (left, right), EMG (Chin1-3), ECG (1 channel)	
MIT-BIH [12]	–	R&K	Patients; 32–56	18	EEG (C4-A1, O2-A1, C3–O1, C4-A1), EOG (1 channel), EMG (Chin)	https://physionet.org/content/slpdb/1.0.0/
SHHS [13,14]	SHHS-1 SHHS-2	R&K	Mixed; >40	6441 3295	EEG (C3-A2, C4-A1), EOG (left, right), EMG (submental), ECG (1 channel)	https://sleepdata.org/datasets/shhs
SIESTA [15]	–	R&K	Mixed; 20–90	295	EEG (Fp1-M2, C3-M2, O1-M2, Fp2-M1, C4-M1, O2-M1), EOG (2 channels), EMG (submentalis, bilateral anterior tibialis), ECG (1 channel)	–
Sleep-EDF [16,17]	SC ST	R&K	Healthy; 21–35 Patients; 21–35	4 4	EEG (Fpz-Cz, Pz-Oz), EOG (horizontal), EMG (submental)	https://physionet.org/content/sleep-edf/1.0.0/
Sleep-EDF-2018 [16,17]	SC	R&K	Healthy; 25–101	153	EEG (Fpz-Cz, Pz-Oz), EOG (horizontal), EMG (submental)	https://physionet.org/content/sleep-edfx/1.0.0/
	ST		Patients; -	44		
UCD [17]	–	R&K	Patients; 28–68, 50 ± 10	25	EEG (C3-A2, C4-A1), EOG (left, right), EMG (submental), ECG (Modified lead V2, with 3-channel Holter ECG [V5, CC5, V5R])	https://physionet.org/content/ucddb/1.0.0/

AASM: American Academy of Sleep Medicine, CAP: Cyclic Alternating Pattern, CinC: Computing in Cardiology Conference, ECG: electrocardiogram, EDF: European Data Format, EEG: electroencephalogram, EMG: electromyogram, EOG: electrooculogram, ISRUC: Institute of Systems and Robotics, University of Coimbra, MASS: Montreal Archive of Sleep Studies, MIT-BIH: Massachusetts Institute of Technology-Beth Israel Hospital, R&K: Rechtschaffen and Kales, S: subgroup, SC: Sleep Cassette, SHHS: Sleep Heart Health Study, SS: sleep study, ST: Sleep Telemetry, UCD: University College Dublin. SIESTA: a new standard for integrating polygraphic sleep recordings into a comprehensive model of human sleep and its validation in sleep disorders.

4.1. Signals

In contrast to visual scoring, which predominantly relies on the EEG, electrooculogram (EOG), and electromyogram (EMG), different types and quantities of physiological signals, along with their combinations, can be used for DL-based sleep architecture analysis.

4.1.1. Single-modal signals

Single-modal signals are physiological signals originating from the same sensor or source. Algorithms based on single-modal signals are preferable because of the simplicity of their hardware implementation. Most methods rely solely on single-channel EEG for automated sleep staging [21–29], owing to the direct reflection of different sleep stages in the changes in brainwave frequencies and rhythms. However, the optimal placement of single-channel electrode remains unclear. Similarly, some studies have employed only single-channel EOG signals for sleep stage classification [30]. Furthermore, research has indicated that autonomic nervous activity varies with sleep, leading to subtle changes in features of electrocardiogram (ECG), including heart rate, R–R intervals, heart rate variability (HRV), and ECG-derived respiration, which provides information about sleep structure [31]. Therefore, researchers have favored using a single-channel ECG for sleep stage identification [32–34].

Notably, the term "single-channel" is commonly used in literature; however, it usually refers to "single lead," where one lead consists of one pair of electrodes. Considering EEG, EOG and EMG, although multiple leads exist, algorithms that use multi-channel signals are still considered within the scope of models with "single-modal signals".

4.1.2. Multi-modal signals

Compared with relying solely on single-modal signals, incorporating multi-modal signals into DL offers the advantage of extracting a richer set of sleep information. The most commonly used multi-modal fusion approach combines EEG with EOG and/or EMG signals [35–39]. EOG is the initial criterion for distinguishing REM sleep, whereas EMG provides a stable indicator of REM sleep. Combining these signals with EEG data enhances the comprehensiveness of sleep structure analysis. ECG combined with body movement is another widely used multi-modal fusion approach that can be applied to healthcare ECG monitoring devices, such as smart hardware and wristbands, with minimal sleep disruption [40–43]. Some studies have proposed exhaustive approaches that utilize multi-modal signals that combine the pattern recognition of sleep stages,

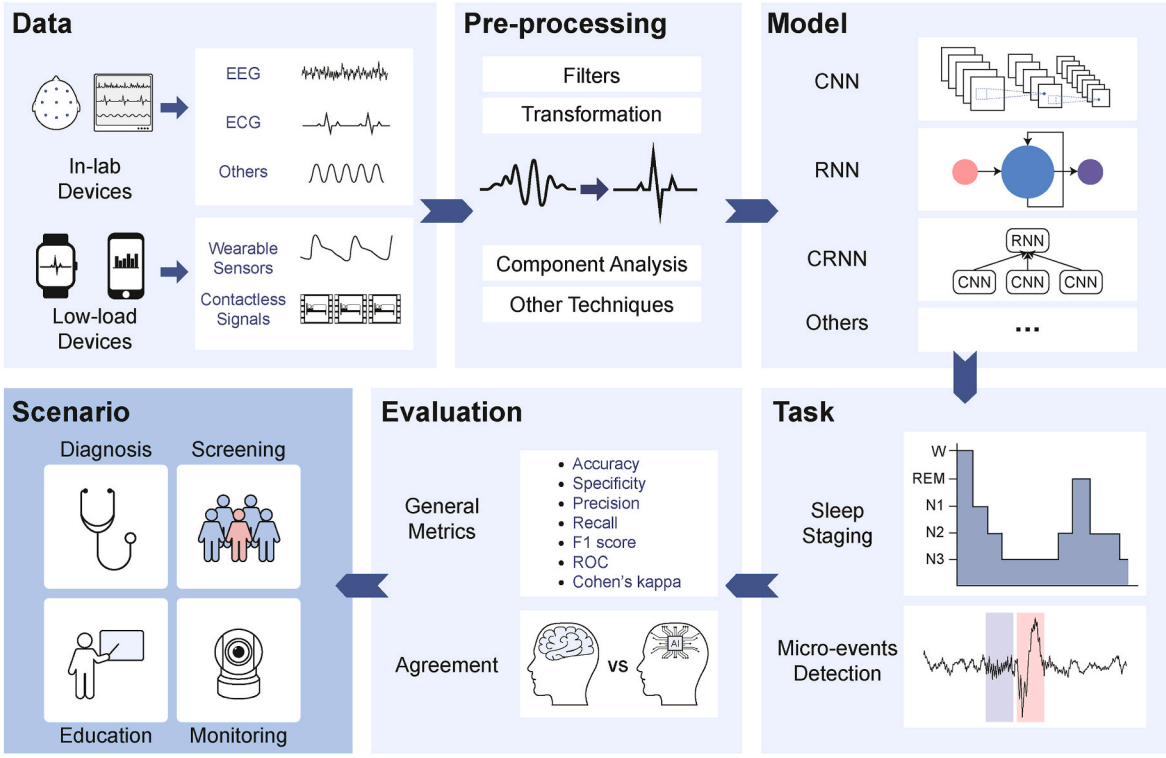

Fig. 1. Overview of analysis based on the aspects of signal selection, data pre-processing, model architecture, classification task, performance evaluation, and application scenario. AI: artificial intelligence, CNN: convolutional neural network, CRNN: convolutional recurrent neural network, ECG: electrocardiogram, EEG: electroencephalogram, N, non-rapid eye movement, REM: rapid-eye movement, RNN: recurrent neural network, ROC: receiver operating characteristic curve, W: wakefulness.

sleep apnea events, limb movements, and other events with AI classifiers [44,45]. Additionally, some fusion models offer the flexible integration of various signals, ensuring they can operate effectively even if any signal is disrupted or lost [46].

Certainly, multi-modal fusion is not always effective, and sometimes, it increases the computational costs and compromises the algorithm efficiency, inevitably conflicting with the goal of relieving the monitoring load. However, given the complexity of sleep and sleep disorders, multi-modal information remains indispensable for rigorous clinical scoring, especially for the medical-grade diagnosis of sleep disorders. Therefore, the primary objective of such studies is to reduce the complexity of fusion models while maintaining the model performance, allowing the incorporation of multiple signal sources in appropriate and available scenarios.

4.1.3. *Other signals*

With continuous innovations in sensor optimization, wireless transmission, and digital signal processing technologies, novel sleep monitoring devices complement each other in signal acquisition, accuracy, and application scenarios, facilitating the transition of sleep monitoring from in-lab to out-of-lab settings, fixed to portable solutions, and restrictive to non-disruptive methods. Therefore, consumer sleep technologies (CSTs) have flourished and are gradually gaining public acceptance [47]. Examples of these innovations include the use of multipin dry electrodes to measure EEG [48] and nanomembrane electrodes to measure EEG, EOG, and EMG [36]. The most common wearable sensor devices are capable of measuring photoplethysmography (PPG) [49] and pulse transit time [50], while passive contact techniques employing piezoelectric sensors, such as sleep mattresses and pillows, can detect both respiratory movements and ballistocardiography [51]. Furthermore, some techniques are based on contactless techniques, such as infrared video [52], audio [53], inaudible sonar [54], and impulse-radio ultra-wideband radar [55], that can capture the status of the entire body and fine respiratory movements. The aforementioned devices are embedded with the corresponding DL algorithms to assess sleep stages. The main challenge with these methods is that the signals from these novel sensors may differ from standard medical metrics, resulting in diagnostic modes of binary classification or risk grading. However, CSTs have significantly improved the convenience and availability of sleep monitoring, facilitating real-time and cost-effective sleep analysis in nonclinical scenarios.

4.2. *Pre-processing*

Raw signals are often contaminated with noise, and pre-processing can reduce noise and interference, thereby making the signals more suitable for specific algorithms and easier to be analyzed. However, some studies aimed at achieving no pre-processing were implemented on raw signals [56,57]. Additionally, some studies have suggested that pre-processing raw signal data may not significantly improve the model's performance; conversely, the model may become more robust when trained on noisy data [58].

4.2.1. *Filters*

Filtering is the most common pre-processing method and the core element of the pre-processing pipeline. Common filtering techniques include high-pass, low-pass, bandpass, and notch filtering. High-pass filtering blocks low-frequency components in the signal below a specified threshold while retaining the high-frequency components [59]. In contrast, low-pass filtering removes high-frequency noise from signals [28]. Bandpass filtering allows signals within a specified frequency range to pass through, which helps eliminate baseline drift and high-frequency noise [60]. Generally, the number, type, and frequency thresholds of the filters can be designed based on the signal characteristics, research objectives, and experience. For example, some studies have employed a combination of filters [39,61].

Table 2
Summary of deep learning-based systems for sleep staging that used single neural network.

Authors	Datasets	Signals (channels)	Models	Tasks	Performance		
					Acc (%)	MF1 (%)	κ
Zhu et al. 2020 [21]	Sleep-EDF	Single-channel EEG (Fpz-Cz)	CNN	5#	93.7	84.5	–
	Sleep-EDFX				82.8	77.8	–
Wang et al. 2020 [22]	Sleep-EDF	Single-channel EEG (Fpz-Cz)	CNN	5#	86.1	53.3	0.57
Khalili et al. 2021 [23]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	CNN	5#	85.4	79.3	0.80
	Sleep-EDF-2018				82.5	76.1	0.76
Eldele et al. 2021 [24]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	CNN	5#	84.4	78.1	0.79
	Sleep-EDF-2018				81.3	75.1	0.74
	SHHS (329 selected)	Single-channel EEG (C4-A1)			84.2	75.3	0.78
Zhang et al. 2022 [25]	Sleep-EDF-2018 (8 selected)	Single-channel EEG (Fpz-Cz/Pz-Oz)	CNN	5#	89.4	72.0	0.78
An et al. 2022 [26]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	CNN	6	89.5	–	–
	Sleep-EDF-2018				91.7	–	–
Zhou et al. 2023 [27]	Sleep-EDF-2018	Single-channel EEG (Fpz-Oz)	CNN	5#	85.8	77.6	0.85
	MASS-SS3	Single-channel EEG (C4-A1)		5	86.4	80.3	0.86
	HSFU	Single-channel EEG (C4-A1)			82.5	75.1	0.82
Goshtasbi et al. 2022 [28]	Sleep-EDF	Single-channel EEG (Fpz-Cz)	CNN	5	84.8	78.8	0.79
	SHHS (120 selected)	Single-channel EEG (C4-A1)			81.3	72.0	0.74
Zhang et al. 2020 [29]	UCD	Single-channel EEG	CNN	5	88.4	–	0.82
	MIT-BIH				87.6	–	0.80
Sridhar et al. 2020 [33]	SHHS	HR	CNN	4	78.0	–	0.67
	MESA				80.0	–	0.69
	CinC2018				72.0	–	0.55
Jia et al. 2020 [68]	MASS-SS3	Multi-channel EEG	CNN	5#	88.9	84.1	0.83
Jia et al. 2021 [69]	ISRUC-S3	Multi-channel EEG	CNN	5#	82.1	80.8	0.77
	MASS-SS3				89.5	85.4	0.84
Sokolovsky et al. 2020 [35]	Sleep-EDF-2013	EEG (Fpz-Cz) + EOG (horizontal)	CNN	5	81.0	72.0	–
Kwon et al. 2022 [36]	ISRUC	EEG (F3-A2, F4-A1) + EOG (LOC-A2, ROC-A1)	CNN	5#	88.9	–	0.85
	Lab dataset				81.5	–	0.73
Zhu et al. 2020 [37]	Sleep-EDFX	EEG (Fpz-Cz, Pz-Oz) + EOG (horizontal) + EMG (submental)	CNN	5	85.8	81.2	0.80
	UCD	EEG (C3-A2, C4-A1) + EOG (left, right) + EMG (submental)			79.4	78.8	0.73
Zhu et al. 2023 [38]	Sleep-EDFX	Single-channel EEG + EOG + EMG	CNN	5#	85.7	79.2	0.85
	MASS-SS3			5	87.5	82.9	0.87
	HSFU			5	81.1	77.9	0.81
Phan et al. 2019 [39]	MASS (SS1-SS5)	EEG (C4-A1) + EOG (ROC-LOC) + EMG (Chin1-Chin2)	RNN	5/5#	87.1	83.3	0.82
Michielli et al. 2019 [70]	Sleep-EDF-2018 (12 selected)	Single-channel EEG (Fpz-Cz)	LSTM	5#	86.7	–	–
Radha et al. 2019 [34]	SIESTA	HRV	LSTM	4	77.0	–	0.61
Fonseca et al. 2020 [40]	SIESTA, Lab dataset, SOMNIA	HRV + acceleration	LSTM	4	75.9	–	0.60
Choi et al. 2020 [41]	Seoul National University Hospital	Respiration + HRV + ballistocardiogram	LSTM	4	73.9	–	0.55
Zhang et al. 2018 [42]	General Hospital of the Air Force	HR	Bi-LSTM	5#	–	53.2	–
		HR + actigraphy			–	58.2	–
Morokuma et al. 2023 [43]	Tokorozawa Respiratory Clinic	HR + respiratory rate + cardiorespiratory coupling + body movement frequency	Bi-LSTM	5#	71.2	65.0	0.43

Acc: accuracy, Bi-LSTM: bidirectional long short-term memory, CinC: Computing in Cardiology Conference, CNN: convolutional neural network, ECG: electrocardiogram, EDF: European Data Format, EDFX: European Data Format expanded, EEG: electroencephalogram, EMG: electromyogram, EOG: electrooculogram, HR: heart rate, HRV: heart rate variability, HSFU: Huashan Hospital Fudan University, ISRUC: Institute of Systems and Robotics, University of Coimbra, LSTM: long short-term memory, MASS: Montreal Archive of Sleep Studies, MESA: Multi-Ethnic Study of Atherosclerosis, MF1: Macro-F1 score, MIT-BIH: Massachusetts Institute of Technology-Beth Israel Hospital, RNN: recurrent neural network, S: subgroup, SHHS: Sleep Heart Health Study, SS: sleep study, UCD: University College Dublin, κ: Cohen's kappa coefficient. 4: W/light sleep (S1+S2)/deep sleep (S3+S4)/REM; 5: W/S1/S2/SWS (S3+S4)/REM; 5#: W/N1/N2/N3/REM; 6: W/S1/S2/S3/S4/REM; SIESTA: a new standard for integrating polygraphic sleep recordings into a comprehensive model of human sleep and its validation in sleep disorders; SOMNIA: the sleep and obstructive sleep apnea monitoring with non-Invasive applications study.

4.2.2. Transformation

Transformation refers to the process of converting a signal from one representation to another to enhance our understanding of its characteristics, structure, and dynamic variations. The wavelet transform (WT) decomposes a signal into sub-bands. Noisy and undesired bands are eliminated by calculating the energy in each sub-band and applying a threshold. In a previous study, a continuous WT was applied to extract musical beat-based features from the EEG [62]. Fourier transform analyzes signal characteristics and components by decomposing the signal into a series of sine and cosine waves of different frequencies [63]. The short-time Fourier transform (STFT) overcomes the limitations of the Fourier transform, which applies only to stationary signals (where the frequency characteristics do not change over time). Some researchers converted each 30-s EEG epoch into spectrograms using STFT to extract features. Other transformations used for automatic sleep stage classification include the fast Fourier transform [64].

4.2.3. Other techniques

Principal component analysis (PCA) is a well-known technique for reducing the dimensionality of multivariate datasets by replacing original variables with new linear combinations, that is, principal components. In a study, PCA was performed on the epoch-wise frequency spectra of EEG to extract principal feature components [65]. Independent component analysis is a process of information unmixing that aims to separate and extract mutually independent components. Chambon et al. used this method to remove EOG signal artifacts [66]. Lajnef et al. [67] used morphometric component analysis to decompose complex EEG signals into morphological and non-morphological components.

Table 3
Summary of deep learning-based systems for sleep staging that used multiple neural networks.

Authors	Datasets	Signals (channels)	Models	Tasks	Performance		
					Acc (%)	MF1 (%)	κ
Fan et al. 2021 [30]	Sleep-EDF-SC	Single-channel EOG (horizontal)	CRNN	5	76.3	69.3	67.2
	MASS-SS3	Single-channel EOG (right)		5#	81.2	73.7	71.8
Supratak et al. 2020 [78]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz/Pz-Oz)	CNN + LSTM	5	85.4	80.5	0.80
	Sleep-EDF-2018				83.1	78.1	0.77
	MASS	Single-channel EEG (F4-EOG/C4-EOG)		5#	87.5	83.2	0.82
He et al. 2022 [60]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	CNN + LSTM	5	83.7	77.7	0.77
	Sleep-EDF-2018				81.6	76.4	0.75
Zhao et al. 2022 [79]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	CNN + LSTM	5#	84.8	79.8	0.79
	Sleep-EDF-2018				82.7	77.2	0.76
	SHHS	Single-channel EEG (C4-A1)			86.4	80.5	0.81
	CAP	Single-channel EEG (C4-A1/C3-A2)			78.8	72.7	0.71
Chen et al. 2021 [80]	Clinical dataset	HRV + acceleration	CNN + LSTM	2	94.5	–	–
Yan et al. 2021 [77]	Sleep-EDF-2013 (SC)	EEG (Fpz-Cz, Pz-Oz) + EOG (horizontal)	2D-CNN + LSTM	5	86.0	–	0.81
	SHHS-1	EEG (C3, C4) + EOG (ROC, LOC) + EMG + ECG			87.0	–	0.81
	ISRUC-S1	EEG (F3, C3, O1, F4, C4, O2) + EOG (ROC, LOC) + EMG + ECG		5#	86.0	–	0.82
Mousavi et al. 2019 [81]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	CNN + Bi-LSTM	5	84.3	79.7	0.79
	Sleep-EDF-2018				80.0	73.6	0.73
Li et al. 2022 [63]	Sleep-EDF	Single-channel EEG (Fpz-Cz)	CNN + Bi-LSTM	5	94.2	87.8	0.91
	Sleep-EDF-2013				86.8	81.6	0.82
	Sleep-EDF-2018				83.0	77.3	0.77
	SHHS (329 selected)	Single-channel EEG (C4-A1)			85.1	78.5	0.79
Korkalainen et al. 2020 [57]	Sleep-EDF-2018	Single-channel EEG (Fpz-Cz)	CNN + Bi-LSTM	5	83.7	–	0.77
		EEG (Fpz-Cz) + EOG (horizontal)			83.9	–	0.78
	Princess Alexandra Hospital	Single-channel EEG (F4-M1)		5#	82.9	–	–
		EEG (F4-M1) + EOG (E1-M2)			83.8	–	–
Sun et al. 2020 [82]	MASS (SS2-SS5)	EEG (C4-A1) + EOG (ROC-LOC) + EMG (Chin1-Chin2)	CNN + Bi-LSTM	5/5#	87.8	81.8	0.82
Sun et al. 2020 [83]	MGH	ECG + ABD	CNN + Bi-LSTM	5#	–	68.1	0.59
Pathak et al. 2021 [59]	SHHS-1	EEG + EOG + EMG (all channels)	CNN + Bi-LSTM	5	85.0	76.6	0.79
	MST			5#	77.2	73.8	0.69
Toma et al. 2023 [84]	Sleep-EDF-2013	EEG (Fpz-Cz, Pz-Oz) + EOG + EMG	CNN + Bi-LSTM	5#	91.4	88.7	0.89
	Sleep-EDF-2018				90.2	87.0	0.86
Li et al. 2023 [85]	Sleep-EDF-2018	Single-channel EEG (Fpz-Cz)	ResNet + LSTM	5#	86.4	82.0	0.81
	ISRUC	Single-channel EEG (F3-A2)		5	82.0	80.0	0.77
Seo et al. 2020 [56]	Sleep-EDF-2013	Single-channel EEG (Fpz-Cz)	ResNet + Bi-LSTM	5	83.6	76.5	0.77
	MASS-SS3	Single-channel EEG (F4-EOG)		5#	86.2	80.0	0.79
	SHHS-1	Single-channel EEG (C4-A1)		5	86.3	78.8	0.81
Liu et al. 2022 [86]	Sleep-EDF-2013	Multi-channel EEG (C3-A2, C4-A1)	ResNet + Bi-GRU	5	84.2	–	0.79
	UCD				79.3	–	–
	SHHS				81.6	–	–
Einizade et al. 2023 [87]	Sleep-EDF-2018	EEG + EOG + EMG (graph structure)	GCN + Bi-GRU	5#	83.8	77.4	0.78
	MASS-SS3			5	86.7	81.8	0.80

Acc: accuracy, ABD: abdominal respiratory effort, Bi-GRU: bidirectional gated recurrent unit, Bi-LSTM: bidirectional long short-term memory, CAP: Cyclic Alternating Pattern, CNN: convolutional neural network, CRNN: convolutional recurrent neural network, ECG: electrocardiogram, EDF: European Data Format, EDFX: European Data Format expanded, EEG: electroencephalogram, EMG: electromyogram, EOG: electrooculogram, GCN: graph convolutional network, HR: heart rate, HRV: heart rate variability, ISRUC: Institute of Systems and Robotics, University of Coimbra, LSTM: long short-term memory, MASS: Montreal Archive of Sleep Studies, MESA: Multi-Ethnic Study of Atherosclerosis, MF1: Macro-F1 score, MGH: Massachusetts General Hospital, MST: Medisch Spectrum Twente, ResNet: residual network, S: subgroup, SC: Sleep Cassette, SHHS: Sleep Heart Health Study, SS: sleep study, UCD: University College Dublin, κ: Cohen's kappa coefficient. 4: W/light sleep (S1+S2)/deep sleep (S3+S4)/REM; 5: W/S1/S2/SWS (S3+S4)/REM; 5#: W/N1/N2/N3/REM; 6: W/S1/S2/S3/S4/REM.

The former represents signals with distinct shapes or features corresponding to different sleep stages, whereas the latter represents noise or other interfering signals. To stabilize the training, Zhu et al. performed z-score normalization on the data from each individual [21].

4.3. Architecture

A summary of methods using CNN and RNN is presented in Table 2.

4.3.1. CNN

CNNs are widely used in image classification, natural language processing, and signal analysis. A CNN architecture comprises four layers: input, convolutional, pooling, and classification (usually a fully connected layer with a softmax function). The convolutional layer is used for feature extraction and consists of multiple feature planes, each composed of neurons. The convolutional and pooling layers are the characteristic structures of CNNs and are often combined into modules stacked on top of each other to form deep networks. Depending on the dimensionality of the input data, CNNs can be further categorized as one-dimensional CNN (1D-CNN), two-dimensional CNN (2D-CNN), and three-dimensional CNN. 1D-CNN is suitable for one-dimensional input data, such as biomedical signals [71]. Yildirim et al. [72] used a 19-layer 1D-CNN model to construct an end-to-end architecture for the classification of sleep stages, whereas Jia et al. [69] implemented a standard 2D convolution layer to extract the temporal context information of the current sleep stage.

4.3.2. RNN

RNNs are neural networks with recurrent connections that perform well in modeling sequential data, such as time series, event sequences, signals, and natural language. A typical RNN considers a sequence as the input; recursion occurs in the direction of the sequence evolution, and the output from the previous step is used as the input for the current step. For sleep-monitoring signals, RNNs are a logical choice for

Table 4
Summary of deep learning-based systems for detection of sleep micro-events.

Authors	Datasets	Signals	Models	Tasks	Performance		
					Pre	Sen	F1
Kaulen et al. 2022 [89]	MASS + MODA	EEG	CNN	Spindle	0.85	0.79	0.82
You et al. 2021 [90]	MASS	EEG	CNN	Spindle	0.86	0.86	0.85
	DREAMS				0.52	0.78	0.54
Chambon et al. 2019 [91]	MASS-SS2 + SSC + WSC + MESA	EEG	CNN	Spindle	0.70	0.77	0.73
				K-complex	0.66	0.57	0.61
Kulkarni et al. 2019 [92]	MASS	EEG	CRNN	Spindle	–	0.90	0.75
	DREAMS				–	0.77	0.48
Tapia et al. 2020 [93]	MASS	EEG	CRNN	Spindle	0.88	0.83	0.85
				K-complex	0.85	0.82	0.83

CNN: convolutional neural network, CRNN: convolutional recurrent neural network, DOSED: dreem one shot event detector, EEG: electroencephalogram, F1: F1 score, MASS: Montreal Archive of Sleep Studies, MESA: Multi-Ethnic Study of Atherosclerosis, MODA: Massive Online Data Annotation, Pre: precision, Sen: sensitivity, SS: sleep study, SSC: Stanford sleep cohort, WSC: Wisconsin sleep cohort. DREAMS: a database from the University of MONS-TCTS Laboratory and Universite Libre de Bruxelles.

capturing temporal dependencies and handling inputs of various lengths. Gated recurrent units (GRUs), long short-term memory (LSTM), and bidirectional LSTM (Bi-LSTM) are commonly-used RNN variants in the reviewed studies. Wang et al. [73] developed a multi-layer graph attention network combining the GRU to capture temporal dependent features and spectral information of the current node. Michielli et al. [70] proposed a cascaded RNN architecture based on LSTM blocks and evaluated its performance on a single-channel EEG. Zhang et al. [42] trained a Bi-LSTM-based RNN model to classify sleep stages using heart rate and wrist actigraphy.

4.3.3. Combination of multiple deep networks

As mentioned previously, different DL models have unique characteristics. Considering combinations of two or more network structures is natural. A convolutional recurrent neural network (CRNN) combines CNN and RNN modules, making it the preferred architecture for handling long physiological signals with varying sequence lengths and multi-channel inputs. In the CRNN architecture, a CNN is used to capture essential features. Thereafter, RNN is employed to learn the sequence relationships between different sleep stages within these features. Subsequently, residual connections are used to fuse different types of features. Finally, a fully connected softmax layer is applied to classify the fused features. Some researchers have proposed a new DL model based on 1D-CNN and LSTM to improve the accuracy of sleep staging [74–76], whereas Yan et al. [77] proposed a new DL model based on 2D-CNN and LSTM. Moreover, some researchers combined CNNs with Bi-LSTM. These results are summarized in Table 3.

4.4. Tasks

Compared to the traditional machine learning, which requires manual feature extraction, DL achieves tasks through end-to-end mapping. In other words, when raw physiological signals are input into the model, the output provides the desired classification or prediction without requiring manual intervention in specific process details. When sufficient data accurately represent the task at hand, DL can automatically discover the best features that match the task through multiple training iterations.

4.4.1. Sleep staging

Different sleep-scoring rules require various automated classification tasks. Models can directly perform six-class classification or five-class tasks based on R&K or AASM rules and can also establish alternative classification types according to their research objectives. For example, a study [88] aimed to detect wakefulness with eyes open, wakefulness with eyes closed, and three sleep stages (N1, N2, and N3), whereas another study primarily designed for sleep-wake detection [80]. For a better comparison with similar studies, researchers usually harmonize the R&K rules with the AASM rules, where S1 and S2 are recorded as N1 and N2, and S3 and S4 are merged into a single stage, N3. Some researchers have focused on differentiating W, S1, S2, SWS (S3+S4), and REM. In some other studies, S1 and S2 were combined to represent NREM light sleep, whereas S3 and S4 were combined to represent NREM deep sleep.

4.4.2. Sleep micro-events detection

During sleep, owing to changes in the internal environment of the body and the excitability of neurons in the cerebral cortex, EEG exhibits various characteristic waveforms, also known as sleep micro-events, such as sleep spindles, K-complexes [89–93], and frequency components, including alpha, beta, delta, and theta activities. Given that these characteristic waveforms play a crucial role in sleep staging and are associated with various types of sleep disorders and mental health conditions, researchers have started using DL to detect and identify them. Kulkarni et al. [92] employed a DL algorithm called SpindleNet to detect sleep spindle waves using single-channel EEG data. Tapia et al. [93] introduced a CRNN model, a recurrent event detector, designed to identify spindle and K-complex events. The recent models for detecting different micro-events using DL are listed in Table 4.

4.5. Performance

Once a new model has been trained and validated, its performance evaluation typically involves using an independent internal test set drawn from the same large dataset as the training and validation sets, with non-overlapping samples based on a specific ratio. To further validate the generalization ability of the model for unknown data, additional datasets independent of the training, validation, and internal test sets are required for external validation.

4.5.1. General metrics

Herein, we summarize the most common parameters used to evaluate the classification performance in previous studies, including accuracy (Acc), specificity (Spc), precision (or positive predictive value, PPV), and recall (or sensitivity, Sen). These parameters can be calculated with the values of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) and can be usually visualized through a confusion matrix. Based on the parameters mentioned above, the F1 score can be calculated. Furthermore, a receiver operating characteristic curve can be created by considering the true positive rate versus the false positive rate with different thresholds for the classifier. The area under the curve can also be calculated. The formula used to calculate the above indicators is as follows:

$$Acc = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (1)$$

$$Spc = \frac{TN}{(TN + FP)} \tag{2}$$

$$PPV = \frac{TP}{(TP + FP)} \tag{3}$$

$$Sen = \frac{TP}{(TP + FN)} \tag{4}$$

$$F_1 = 2\frac{PPV * Sen}{PPV + Sen} \tag{5}$$

To comprehensively assess the performance in the sleep-staging task, some authors considered the model evaluation of both the overall and per-category separately. Overall Acc, Macro-F1 score [94], and Cohen's kappa coefficient (κ) were used to obtain a more intuitive and realistic portrayal of the overall classification. κ is a measure of agreement between the model and manual labels, considering the probability of agreement occurring by chance, which is a more robust measure than Acc when dealing with imbalanced datasets [95].

Remarkably, the performance evaluation of an algorithm should also consider the characteristics of the dataset and the classification task, and the validation and testing methods employed. Furthermore, owing to the wide range of applications for automatic sleep staging, there is no one-size-fits-all threshold for acceptable algorithm accuracy that can be adjusted based on specific circumstances. For example, when used in assisting clinical diagnosis, the minimum accuracy threshold may be set to >95 %, whereas for commercial health-monitoring products, an accuracy percentage of 80 % could be considered reasonable. Moreover, when the algorithm achieves satisfactory accuracy, an algorithm with high Sen and low Spc is more suitable for disease screening than diagnosis.

4.5.2. Agreement between automatic and visual scoring results

A significant portion of the work concludes after evaluating the abovementioned performance metrics, which are insufficient for integrating the model into clinical applications. Researchers are increasingly emphasizing the interpretation of each record in practice, translating DL insights into individual segments of the signal into actionable medical recommendations. The common practice is to use human expert visual scoring as the gold standard to evaluate the inter-rater reliability between automatic and expert scorers. This involves related statistical approaches, such as κ, intraclass correlation coefficient (ICC) [96], and Bland-Altman plots for visualizing consistency [97]. Here, κ is a measure of agreement between the proposed model and a human expert.

5. Application scenario

The demand for sleep health is becoming increasingly diverse and multifaceted. Further expansion of automatic sleep-staging application holds promise for catalyzing the transition from traditional healthcare services to digital healthcare and ultimately towards intelligent healthcare, establishing a reliable starting point in enhancing global sleep health.

5.1. Diagnosis

Obstructive sleep apnea (OSA) is a complex and heterogeneous disorder with significant differences among the different phenotypes regarding pathogenesis, clinical presentation, susceptibility to comorbidities, and treatment response. Some studies focused on sleep stages (such as REM sleep predominance) and sleep fragmentation have captured the heterogeneity in OSA. Therefore, the accurate identification of sleep stages is essential for diagnosing OSA. Korkalainen et al. [57] developed a reliable, fast, and accurate DL-based automatic method for classifying sleep stages in patients with suspected OSA. Huttunen et al. [98] developed a DL model to simultaneously score both sleep stages and respiratory events to estimate the apnea-hypopnea index (AHI) (ICC = 0.946) and REM-AHI (ICC = 0.912).

Similarly, owing to the high heterogeneity of insomnia, sleep analysis is crucial for the etiological detection and differential diagnosis. Shahin et al. [99,100] proposed sleep stage-dependent discrimination in two studies to automatically detect insomnia. In the first stage, they trained a sleep-stage scoring model and an epoch-level insomnia detection model. In the second stage, the features obtained from the first stage were used to train a final binary classifier to classify each subject as a control, that is, with no sleep complaints or insomnia. The F1 score, Sen, and Spc were 0.88, 84 %, and 91 %, respectively.

REM sleep behavior disorder (RBD) is a parasomnia characterized by dream enactment, abnormal jerks, and movements during REM sleep, affecting up to 70 % of patients with Parkinson's disease (PD). Confirmation of abnormal behavior during video monitoring using polysomnography (PSG) and REM sleep without atonia based on physiological indicators are essential diagnostic criteria for RBD. Cesari et al. [101] developed an automated system to identify the RBD in patients with PD using only EEG and EOG signals. They applied a previously developed and validated automated sleep-staging algorithm to perform both macro (30-s epochs) and micro (5-s mini-epochs) sleep staging [102]. For each participant, features describing the micro-sleep structure, EEG spectral content, coherence, and complexity and EOG energy were calculated for certain sleep epochs. Therefore, the system identified RBD with an Acc, Sen, and Spc of >80 %.

Narcolepsy is a well-characterized set of sleep disturbances that include short sleep latency, rapid transitions into REM sleep, and poor nocturnal sleep consolidation, and can be classified into narcolepsy type 1 (NT1) and type 2. Stephansen et al. [103] first developed an automatic sleep-staging model based on a combination of CNN and LSTM. The model's results are displayed as hypnodensity graphs, which represent discrete sleep-stage outputs and the probability of occurrence of each sleep state for each epoch. Subsequently, feature extraction was performed based on the hypnodensity graph to identify NT1, achieving 96 % Spc and 91 % Sen. Shen et al. [104] probed into the amalgamation of automatic sleep staging and sleep transition dynamics for narcolepsy detection, which can efficiently distinguish patients with narcolepsy based on multiple sleep latency tests.

5.2. Screening

In practice, the primary diagnosis of OSA is increasingly reliant on home sleep apnea tests (HSAT). However, respiratory-only monitoring does not allow the diagnosis of hypopnea with arousal without significant desaturation. Furthermore, the absence of a measure of total sleep time (TST) also leads to variable underestimation of the AHI because the denominator is replaced by the total recording time instead of TST. Identifying sleep stages based on non-EEG signals could significantly improve the OSA screening yield with minimal disruption to sleep. Korkalainen et al. [105] developed an automatic method using only PPG signals to derive commonly used sleep parameters, such as TST and sleep efficiency. Strumpf et al. [106] assessed the performance of Belun Ring, a class II HSAT with second-generation algorithms in adults suspected of having OSA, and achieved accuracies of 0.88, 0.82, and 0.90 in detecting W, NREM, and REM sleep, respectively. Notably, for patients with a high propensity for arousal, at-home sleep staging may be more representative than laboratory PSGs.

5.3. Health monitoring and management

Unlike diagnostic and screening approaches that prioritize "higher accuracy and more comprehensiveness," monitoring and management focus on "less disturbance and more convenience," which facilitates continuous monitoring and early warning at the household and community levels. In this context, the lightweight networks show greater

potential, which occupy less memory and have a faster inference speed, making them easier to apply to mobile phones or wearable devices [107]. These systems may not be able to identify each individual sleep stage; however, they have shown promising performance in distinguishing wakefulness from sleep [108], offering regular and personalized sleep assessment reports to each user.

6. Discussion

First, obtaining large-scale sleep data in real-world environments remains a major challenge. Fortunately, some novel data augmentation strategies have been developed to address this issue. To alleviate the class-imbalance problem caused by the original distribution of different sleep stages, a focal loss function was introduced into the CNN classifier [22], and four data augmentation methods were designed [60]. Moreover, Mousavi et al. [81] constructed SleepEEGNet using the synthetic minority over-sampling technique. To overcome the overfitting problem, Khalili et al. [23] employed a new data augmentation method called the Mapping Label, which can improve the accuracy by 1.7 % compared with the baseline method without augmentation techniques. Supratak et al. [78] developed a novel training technique equipped with "signal and sequence augmentation," enhancing robustness of the model to new patterns.

"Big data" implies more than a large quantity of data. The high heterogeneity in human sleep and manual PSG scoring makes the diversity in data characteristics more significant. As expected, training the classifier on healthy individuals and validating it on patients with sleep disorders often fails [109]. Ideally, the database should include data from multiple centers worldwide to reduce biases caused by data-related factors, such as race, health status, monitoring techniques, and sensing devices. Building such a large-scale data platform was unimaginable in the past; however, contemporary information technologies, such as 5G networks, Internet of Things, and cloud computing, have made it feasible. This may facilitate international and interdisciplinary cooperation in sleep medicine, sparking innovative research. Furthermore, granting public access to these databases is a step towards open science; however, thoroughly exploring issues related to data security, privacy, ethics, and legal considerations before sharing is vital.

Second, deep supervised learning requires reliable annotated data to achieve higher accuracy and better generalization. However, in the health domain, where the labeling process is particularly costly and the clinical ground truth is often challenging to define, obtaining large quantities of sleep-expert annotations can be difficult. To remove the dependency on labeled data at the representation extraction stage in traditional supervised learning methods, semi-supervised [110], unsupervised [111], and self-supervised learning [112] are applied in sleep staging. This allows the training of more powerful models, with insufficient labeled data and even without labeled data. Li et al. [113] proposed a Bi-Stream Adversarial Learning network to generate pseudo-labels with higher confidence for semi-supervised method optimization and obtained comparable performance with state-of-the-art supervised approaches. Recently, researchers have found that TL can effectively address the problem of model performance degradation due to data distribution bias. This can be achieved through various methods, such as model transferability [77], fine-tuning strategies [114], retrain transfer [49], knowledge transfer [115], and domain adaptation [110]. By adopting two techniques for training (one that transfers learning from hospital data to home audio, and another that ensures consistency in the presence of home noise), the model proposed in a previous study can accurately predict sleep stages using home sounds [116]. Waters et al. [117] indicated that re-training of the head layers is the most effective in most cases and transferability measures are useful indicators of TL effectiveness.

Thirdly, pursuit of high-performance DL models has also catalyzed other research directions, such as more general multi-modal data, including text, images, and other types of data, along with the development of corresponding algorithms to achieve better results [3]. Moreover, the complexity of sleep disorders sometimes requires the interpretation of sleep stage-related events, leading to increased multi-tasking learning approaches in the field. Another important requirement for automatic sleep analysis is real-time processing, which can expedite adjustments to treatment plans. Providing a clear explanation of the results of the DL models is fundamental for helping clinical professionals understand and trust the results of automatic sleep staging. Although some studies have used confidence-based frameworks to differentiate between reliable and possibly incorrect predictions, which can avoid manually reviewing the entire PSGs, the underlying principles behind DL model predictions remain unknown. Therefore, the decision process of automatic scoring systems should be transparent and traceable, and the presentation of the results should be visualized. Additionally, achieving precise localization of sleep stages, including onset time, frequency, and duration, is important.

Furthermore, current DL algorithms break down sleep into fixed-length time segments for identification, which may not accurately assess epoch transitions. This is likely influenced by the AASM visual scoring rules. However, in a clinical setting, scorers naturally observe the transitions between epochs and can also incorporate their knowledge and experience to mitigate the negative impact of "segmentation" on the assessment. With the introduction of continuous digital markers, such as power spectral analysis [118], and a shift towards micro-feature extraction, such as cyclic alternating patterns [119], DL may closely mimic the human reasoning process and even have the potential to uncover important information that has been overlooked by visual scoring.

In addition to the data and technical challenges mentioned earlier, translational research on automatic sleep staging must be further developed to accelerate the validation and application of trained DL models in patients with sleep disorder [120]. Although many researchers have reported the high accuracy of their proposed automatic sleep-staging methods, the lack of validation with large samples in real-world environments and prospective assessments leaves their clinical utility uncertain. Establishing a scientifically sound validation framework applicable to different levels of sleep scoring centers is crucial to ensure the robustness of DL algorithms before their implementation in clinical practice. Additionally, just like human practitioners require continuous learning over time, automatic sleep-staging systems can continuously improve their long-term generalizability through re-training and adaptation based on prospective validation results.

Currently, teams developing sleep-staging methods based on DL primarily consist of AI scientists and algorithm engineers, which is insufficient. Model developers may be confined to training with public datasets and laboratory validations without a precise understanding of the clinical basis and significance of their work. Sleep professionals may identify specific clinical issues related to sleep disorders without suitable approaches to solve these problems. In the future, using a mature interdisciplinary research framework, the same questions can be addressed from different perspectives by experts in sleep medicine, AI theorists, DL engineers, and human factor engineering scientists [121].

Although fully automated sleep staging is the ideal choice, clinical professionals remain the crucial final step in implementing medical decisions, playing an important supervisory role for safety. The organic integration of DL software with human intervention represents an improved medical service model, user-centered usability will be a key area of focus. Many software solutions in the market lack user-friendly interfaces, which may hinder the timely assessment of medical conditions. Therefore, optimizing and enhancing the user experience of automatic sleep-staging systems through multi-modal human-computer interaction remains crucial. This will ensure high-quality and efficient information input and output and promote effective and long-term communication between users and software, ultimately paving the way for more interactive sleep interventions [47].

Practice points

1. Utilizing DL for the automatic analysis of multiple physiological signals obtained from sleep monitor can achieve efficient and accurate sleep staging and sleep micro-events detection.
2. CNN, RNN, and CRNN are the three most commonly used DL architectures. Because their respective advantages and the lack of performance comparisons on the same dataset, it is difficult to evaluate which of them is the optimal; perhaps designing architectures based on the actual requirements of application scenarios is reasonable and preferable.
3. Single-modal models help improve sleep quality; multi-modal fusion models are indispensable for medical-grade diagnosis of sleep disorders; and DL models based on novel sensor signals enhance the non-intrusive sleep assessment methods.
4. Sleep service providers need to understand the mechanisms underlying DL-related recommendations, and it is essential to approach automatic scoring results with professional knowledge and experience.

Research agenda

1. Prepare high-quality annotations, high heterogeneity, and multi-modal data to enhance reliability while achieving interpretability for easy human intervention.
2. Assemble a multidisciplinary team and scientifically validate the new sleep staging program to facilitate the transition of deep learning models from "in-lab" to "real-world" application.
3. Clearly differentiate between clinical utility, research and education, and commercial applications based on different service recipients, job tasks, available time, hardware, and software resources to develop sleep staging applications and enhance the human-computer interaction experience.
4. With the integration of various sleep monitoring technologies and the application of intelligent technologies becoming a new exploration direction, there is hope to see the presence of automatic sleep staging in future guidelines and standards.

The final and crucial aspect is the integration of automatic sleep-staging systems into professional education and training programs. Recently, the New England Journal of Medicine emphasized in its "Perspective" column that curriculum designers, physician training programs, healthcare institution leaders, and certifying bodies must begin actively incorporating AI into physician training [122]. It is known that becoming an experienced registered polysomnographic technologist is a lengthy and challenging journey. Since automated scoring systems have the potential to rival sleep experts, beginners can certainly use DL results to identify errors, investigate the reasons behind them. Developing interactive educational software for them to practice repeatedly makes training methods more efficient, engaging, and beneficial for achieving standardized levels among learners.

7. Conclusion

DL has emerged as a cutting-edge approach and core technology in the field of clinical sleep medicine and fundamental sleep research, offering significant assistance; however, it has also faced some unresolved challenges and issues. Future work should focus on building a sleep big data platform to establish standards for annotations among different studies, developing universal and interpretable models, subjecting them to rigorous testing and extensive validation, promoting interdisciplinary collaboration, optimizing software user interfaces, and integrating professional education and training. Taken together, the journey from research and development to effective integration into routine applications for automatic sleep staging remains challenging, and objective assessment cannot replace comprehensive clinical evaluation. We believe that the inherent accuracy and consistency, efficiency, and cost-effectiveness of DL methods are just the beginning of the "surprises" they bring. We look forward to the next generation of AI opening new doors to explore sleep disorders, thereby advancing sleep science for the health and well-being of patients.

Conflicts of interest

The authors do not have any conflicts of interest to disclose.

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